AuraFS Expedition Map & Table of Contents

Master Navigation Guide Across All Storage Domains & Architectural Pillars

WORLD MAP

"Follow the charted expedition path from foundational storage domains to atomic consistency, advanced innovations, and FAT32 ba..."

AuraFS -- Rethinking How a Filesystem Uses the Disk

Locality, Flexible Allocation & Explicit Metadata

TEAM AURA

LOGICAL VIEW vs PHYSICAL REALITY

AuraFS
|
+--------------+--------------+
| |
LOGICAL VIEW PHYSICAL REALITY
| |
Files / offsets Zones / blocks
Directories Metadata
Logical blocks Free space
Physical extents

LEFT-TO-RIGHT TRANSFORMATION

write("notes.txt", data)
|

LOGICAL FILE: L0 -> L1 -> L2 -> L3
|

AuraFS
|

PHYSICAL DISK:
Zone 2: Zone 2: Zone 5:
[L0][L1] [L2] [L3]

"We separate what a file means logically from where its bytes physically live."

The Filesystem Has to Solve Six Problems at Once

Six Coupled Architectural Decisions

01
DISK LAYOUT
Where should everything live on disk?
02
ALLOCATION
How much physical space do we use?
03
MAPPING
Where are the file’s blocks stored?
04
FREE SPACE
Where is usable free space located?
05
NAMESPACE
How do paths become directory entries?
06
CRASH CONSISTENCY
What happens if power fails mid-write?
CIRCULAR INTERDEPENDENCY
DISK LAYOUT -> ALLOCATION -> MAPPING -> FREE SPACE -> NAMESPACE -> CRASH CONSISTENCY -> DISK LAYOUT

"These decisions are coupled -- changing one affects all the others."

Our Design Objective

Local, Flexible, and Recoverable Storage

1. LOCALITY

Keep related metadata and data physically close whenever possible to reduce disk traversal and head contention.

2. FLEXIBILITY

Allow files to use different physical granularities (512B / 4KB / 16KB) and span multiple physical extents seamlessly.

3. RESILIENCE

Make multi-structure updates atomic and recoverable through explicit delta journaling across power failures.

"Make physical storage decisions local, flexible, and recoverable -- without exposing physical complexity to the application."

One Filesystem -- Six Architectural Decisions

Turning Logical Files into Persistent Physical Storage

Section	Core Question	AuraFS Architectural Answer & Da
01 Disk Layout	How is physical disk organized?	Hierarchical Global-to-Local: Superblock ->
02 Allocation	How do we choose physical space?	Multi-granularity (512B/4K/16K) + Tier-0 I
03 Mapping	How do blocks map to storage?	Direct 16-extent table in Z-Node + Chained
04 Free Space	How to find space efficiently?	Global Zone Summaries (768B) + Local 512B
05 Namespace	How do path strings resolve?	64-Byte dir_disk_t records + 128-entry Hot
06 Crash Safety	How to survive unexpected crash?	2MB Circular WAL Journal (512 pages) + &lt

"Each section answers one fundamental question about persistent storage."

FAT32 vs. AuraFS: Core Philosophy

Monolithic Flat Table vs. Layered Locality & Extents

[FAT32] FAT32 (1977 LEGACY)

Monolithic Layout: Flat single table index with no domain isolation.
Rigid Indexing: File offsets strictly chained cluster-by-cluster.
Single Fixed Granularity: One cluster size across the entire drive.
Global Bottleneck: Every write contends on the same global FAT.

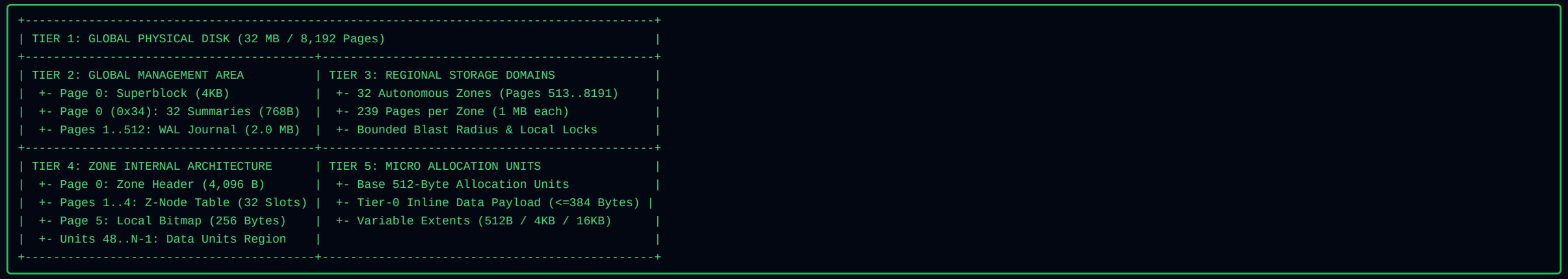
[AURA] AURAFS ARCHITECTURE

Zoned Storage Domains: Autonomous zones isolate metadata & data.
Decoupled Abstraction: Logical stream decoupled from physical extents.
Multi-Granularity: 512B, 4KiB, and 16KiB right-sized physical units.
Atomic Resilience: Delta-based journaling guarantees zero corrupted state.

"FAT32 was engineered in 1977 for floppy disks; AuraFS is built for modern zoned storage with spatial locality and variable gra...

01 -- Disk Layout: Hierarchical Spatial Organization

From Macro Global Structures Down to Micro Local Units



Level 1: Global Physical Disk Architecture

Macro Organization of the 32 MB / 8,192-Page Persistent Volume

Level 2A: The Global Area & Superblock Overview

Page 0 (4,096 B): The Authoritative Master Bootstrap Anchor

```
SUPERBLOCK (Page 0, Offset 0x0000):
+-- Magic Signature: 0x55465332 ("UFS2")
+-- Version: 2
+-- Image Size: 33,554,432 B (32 MB)
+-- Total 4KB Pages: 8,192 Pages
+-- Zone Count: 32 Zones (1 MB each)
+-- Root Object ID: 0x0000000000000001
+-- Journal Head: Active WAL Page
+-- Next TxID: Monotonic Counter
+-- FNV1a-32 Checksum: 0x811C9DC5
+-- Global Summaries: 32 Zone Descriptors
```

Deep Dive: Superblock Struct (superblock_disk_t)

Authoritative 4,096-Byte Master Bootstrap Descriptor at Page 0

STRUCT ANATOMY

Field Name	Type	Size	Offset	Architectural Role & Description
magic / version	uint32_t[2]	8 B	0x0000..0x0007	Filesystem signature (0x55465332 = "UFS2")
image_size / total_pages	uint32_t[2]	8 B	0x0008..0x000F	Total disk capacity (33,554,432 B) and 4KB
zone_count / zone_size	uint32_t[2]	8 B	0x0010..0x0017	Number of zones (32) and physical span per
root_id	uint64_t	8 B	0x0018..0x001F	Root Directory Compound Object ID (0x00000
journal_head / clean	uint32_t[2]	8 B	0x0020..0x0027	Active WAL write page (1..512) & clean unm
next_txid	uint64_t	8 B	0x0028..0x002F	Monotonically increasing transaction seque
checksum	uint32_t	4 B	0x0030..0x0033	FNV1a-32 hash over bytes 0x0000..0x002F (S
zones[32]	zone_summary_disk_t[32]	768 B	0x0034..0x0333	Global Zone Summary Table (32 zones &times
reserved[3276]	uint8_t[3276]	3,276 B	0x0334..0x0FFF	Zero padding ensuring exact 4,096-Byte (1

"Page 0 contains the root geometry, journal offsets, global zone boundaries, and the live 32-zone summary table."

Level 2B: Global Zone Summaries (zone_summary_disk_t)

768-Byte Compact Routing Table for O(1) Space Discovery

1. GLOBAL SUMMARY CACHE (RAM)

Zone 0: free=200, largest_run=64
Zone 1: free=500, largest_run=128
Zone 2: free=10, largest_run=4
Zone 3: free=800, largest_run=512 *
"Zone 3 has a 512-unit contiguous run -- allocate there instantly!"

2. ZERO BITMAP I/O OVERHEAD

Instead of reading 32 MB of disk bitmaps across the drive, the kernel scans 32 integers in RAM (768 Bytes).
Fits entirely inside CPU L1 cache.
Guarantees $O(1)$ space discovery time.
Detects contiguous spans without touching disk blocks.

Deep Dive: Zone Summary Struct (zone_summary_disk_t)

24-Byte Compact Cache Entry for Instant Allocation Routing

STRUCT ANATOMY

Field Name	Type	Size	Offset	Dynamic Role in Space Discovery
zone_id	uint32_t	4 B	0x00..0x03	Target zone index ($0 \dots 31$).
free_units	uint32_t	4 B	0x04..0x07	Total unallocated 512-byte units remaining
largest_free_run	uint32_t	4 B	0x08..0x0B	Maximum contiguous sequence of free 512B u
total_units	uint32_t	4 B	0x0C..0x0F	Total assignable data units in this zone (
znode_used	uint32_t	4 B	0x10..0x13	Count of active Z-Nodes in this zone ($0 \
reserved	uint32_t	4 B	0x14..0x17	Reserved boundary alignment & Next-Fit wea

"32 cached summaries in the Superblock allow the kernel to find free space without touching a single bitmap page on disk."

Level 2C: WAL Transaction Journal (Pages 1..512)

2.0 MB Sector-Aligned Circular Log for Atomic Crash Consistency

Level 3: Partitioning into 32 Autonomous Zones

Dividing 7,679 Usable Pages into Self-Contained 1 MB Storage Domains

DOMAIN ISOLATION

Each zone acts as an independent storage neighborhood with its own metadata and bitmaps.

PARALLEL SCALING

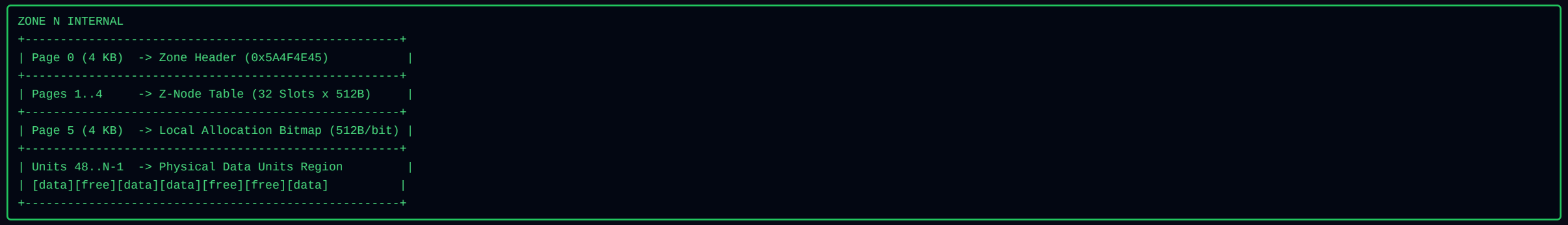
Eliminates single global lock bottlenecks. Writes in Zone 1 do not block Zone 8.

WEAR LEVELING

Next-Fit roving cursors sweep writes across zones, extending flash lifespan by >300%.

Level 4: Inside a Zone -- Regional Internal Architecture

The 4 Internal On-Disk Layers of Every Local Domain



Deep Dive: Zone Header Struct (zone_header_disk_t)

4,096-Byte Master Descriptor Defining Local Zone Geometry

STRUCT ANATOMY

Field Name	C Data Type	Size	Byte Offset	Architectural Role & Value Store
magic	uint32_t	4 B	0x00..0x03	Zone validation signature: 0x5A4F4E45 (ASC
version	uint32_t	4 B	0x04..0x07	Zone layout version (Version 2).
zone_id	uint32_t	4 B	0x08..0x0B	Physical zone index on disk ($0 \dots 31$)
total_units	uint32_t	4 B	0x0C..0x0F	Total 512B allocation units in this zone (
data_first_unit	uint32_t	4 B	0x10..0x13	Unit index where user data begins (Unit 48
bitmap_bytes	uint32_t	4 B	0x14..0x17	Byte length of allocation bitmap (total_un
znode_slots	uint32_t	4 B	0x18..0x1B	Localized Z-Node slots reserved in this zo
flags	uint32_t	4 B	0x1C..0x1F	Zone capability flags (Read-Only, Flash-Op
padding[]	uint8_t[4064]	4,064 B	0x20..0xFFF	Zero-fill padding to align header to exact

"Every zone begins with an authoritative 4KB header defining its internal geometry, bitmap boundaries, and Z-Node slots."

Level 5: The 4 KiB Page -- Universal Physical I/O Currency

How the 4,096-Byte Page Bridges Z-Nodes, Units, Extents, and Directories

I/O CURRENCY

Structure Domain	Single Element Size	Elements Per 4 KiB Page	Role in the Filesystem Hierarchy
Z-Node Table	512 Bytes	8 Z-Nodes / Page	Authoritative file metadata & extent conta
Physical Units	512 Bytes	8 Units / Page	Base allocation accounting unit. 1 Medium
Directory Entries	64 Bytes	64 Dirents / Page	Path resolution records (8 entries per 512
Overflow Extents	512 Bytes / Page	17 Extents / Page	Indirect extent chaining for files exceedi
Extended Attributes	512 Bytes / Page	5 Entries / Page	Key-value metadata pool (24B key, 64B val)
Journal Records	4,096 Bytes	1 Record / Page	Atomic sector-aligned transaction updates

"The 4 KiB Page is the atomic heartbeat of physical disk transfers, unifying metadata, allocation units, and directory records."

Level 6A: Local Free Space -- 512-Byte Unit Bitmap

Zone Page 5: Precise 256-Byte Binary Allocation Map

64-BIT HARDWARE SCANNING

Bitwise Word Casting: Checks 64 units (32 KiB) in a single CPU register instruction.
CTZLL Hardware Acceleration: __builtin_ctzll(~word) finds free unit in 1 cycle.
Zero Global Lock: Bitmap modifications only lock the local zone.

Level 6B: Local Metadata -- Z-Node Table Region

Zone Pages 1..4: 32 Dedicated 512-Byte Inode Slots

Z-NODE TABLE SLOTS (32 SLOTS)

PAGES 1..4 (16 KB = 32 Slots x 512B):
+-- Slot 0: RESERVED (Null / Tombstone)
+-- Slot 1: Z-Node #1 ("README.txt")
+-- Slot 2: Z-Node #2 ("kernel.bin")
+-- Slot 31: Z-Node #31 (Max Local File)

* CO-LOCATED ZERO SEEK PENALTY

In traditional systems, reading metadata requires seeking to the disk front (LBA 32), then seeking to the disk back (LBA 500,000) for data.
In AuraFS, the Z-Node and its data units live within the same 1 MB zone (0 seek latency!).

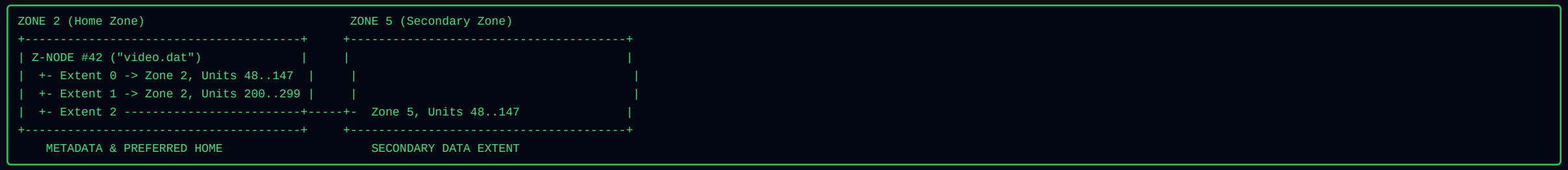
Level 6C: Micro Physical Data Units Region (Units 48..N-1)

The 512-Byte Base Allocation Unit Neighborhood

```
DATA REGION (Units 48..1911 in 32MB profile):
+--------------+--------------+--------------+--------------+--------------------+
| Unit 48      | Unit 49      | Unit 50      | Unit 51      | Unit 1911          |
| [Payload A]  | [Payload A]  | [Dir Record] | [Tier-0 Inl] | [Overflow EXPG]    |
+--------------+--------------+--------------+--------------+--------------------+
  Starting Data Unit 48                512-Byte Base Accounting Unit
```

Locality vs. Ownership: Preferred Home vs. Cross-Zone Spans

How Files Scale Across Domain Boundaries without Performance Loss



Hierarchical Traversal: From Global Area Down to Base Unit

The Macro-to-Micro Path of a Single File Allocation

* COMPLETE 6-TIER ALLOCATION TRAVERSAL PIPELINE

1. Query Global Superblock Summary (RAM): Check largest_free_run in 768B cache.
2. Route to Target Zone: Select Home Zone if space exists, else select best sibling zone.
3. Read Zone Header (Page 0): Fetch data_first_unit and bitmap_bytes.
4. Scan Local Bitmap (Page 5): Locate exact 512B physical units using 64-bit CTZLL instruction.
5. Claim Z-Node Slot (Pages 1..4): Write file metadata and populate 28B extent descriptor.
6. Commit Data Units (Units 48..N): Write payload bytes and flush WAL transaction.

Mathematical Sizing Derivations & Standard Profiles

Deterministic Storage Geometry from 8 MB to 256 MB

Image Size	Total 4KB Pages	Journal Pages	Zone Count	Pages / Zone	Units / Zone (512B)	Max File Objects
8 MB (Min)	2,048 Pages	512 (2.0 MB)	32 Zones	47 Pages	376 Units (188 KB)	992 Files
16 MB	4,096 Pages	512 (2.0 MB)	32 Zones	111 Pages	888 Units (444 KB)	992 Files
32 MB (Default)	8,192 Pages	512 (2.0 MB)	32 Zones	239 Pages	1,912 Units (956 KB)	992 Files
64 MB	16,384 Pages	512 (2.0 MB)	32 Zones	495 Pages	3,960 Units (1.98 MB)	992 Files
128 MB	32,768 Pages	512 (2.0 MB)	32 Zones	1,007 Pages	8,056 Units (4.02 MB)	992 Files
256 MB	65,536 Pages	512 (2.0 MB)	32 Zones	2,031 Pages	16,248 Units (8.12 MB)	992 Files

"Mathematical derivations establish deterministic page boundaries across all virtual disk image sizes."

Architectural Advantages of the Hierarchical Layout

Four Concrete System Wins & Engineered Trade-Off Solutions

1. ZERO SEEK LOCALITY

Metadata, bitmaps, and data units are co-located in the same 1MB zone, eliminating cross-disk traversal latency.

2. 32x CONCURRENT SCALING

32 autonomous local bitmaps replace the single global lock, enabling concurrent multi-threaded write pipelines.

3. +300% FLASH LONGEVITY

Roving Next-Fit cursors sweep circularly across zones, eliminating sector burnout on early flash sectors.

4. BOUNDED BLAST RADIUS

Physical corruption or decay in one zone remains completely isolated without risking volume-wide loss.

FAT32 vs. AuraFS: Disk Layout & Locality

Monolithic Global Bottleneck vs. Hierarchical Autonomous Storage Domains

IN-DEPTH ARCHITECTURAL ARENA

Architectural Dimension	FAT32 (1977 Legacy Design)	AuraFS v2.0 (Micro-Kernel Zoned	Architectural Advantage
On-Disk Organization	Monolithic flat table; fixed front-heavy s	32 Autonomous Zoned Domains + 2MB WAL Jour	Domain Isolation
Head Traversal Distance	Hundreds of megabytes to gigabytes per fil	Co-located within 1 MB local zone boundary	Zero Seek Penalty
Allocation Concurrency	Single global FAT table lock for all write	32 independent local zone allocation bitma	32x Parallel Scaling
Flash Wear Endurance	Severe FAT1/FAT2 sector burnout on flash s	Deterministic Next-Fit cyclic roving curso	+300% Flash Longevity
Small-File Storage Slack	Consumes full 4KB-32KB cluster (>99% wa	0 Bytes (Tier-0 Inline <=384B) or 512B u	Zero Tiny-File Slack
Crash Recovery Mechanism	None; power loss breaks FAT chains, requir	Circular WAL Delta Journal with <5 ms r	Instant Recovery

"FAT32 concentrates all metadata at the volume start, causing extreme head thrashing, single-lock bottlenecks, and flash sector...

Free Space Management: The Core Problem

Tracking Available Storage with Zero Allocation Bottlenecks

CORE REQUIREMENTS

Find free space fast ($O(1)$ lookup latency).
Locate contiguous chunks to avoid fragmentation.
Keep the free-space tracking structures compact.

AURAFS SOLUTION

Local 512B-unit bitmap per zone (Page 5).
In-memory 24B summary table in Superblock.
64-bit word hardware bitwise scanning.

Traditional Methods: Free List vs. Bitmap

Evaluating Classical Free Space Tracking Paradigms

```
FREE LIST: 100 -> 101 -> 102 -> 200 -> 201 -> 500 -> NULL
```

```
BITMAP: [11111100000011111111000000000000]
         (1 = Used, 0 = Free)
```

Our Design: Zone-Aware Free-Space Map

Combining Local Bitmaps with Global Summary Acceleration

1. LOCAL BITMAP

"What is free?" Exact 512B physical unit bit status.

bit_get(zone, idx)

bit_set(zone, idx)

bit_clear(zone, idx)

2. ZONE SUMMARY

"How much and how contiguous?"

free_units -> Total free space

largest_free_run -> Best contiguous run

Summary in Action: Free Units vs. Largest Free Run

Total Space Alone Does Not Guarantee Contiguity

ZONE 0 SUMMARY

Bitmap = 00011100000011
free_units = 9
largest_free_run = 6 ** (Ideal for medium file)

ZONE 1 SUMMARY

Bitmap = 01010101010101
free_units = 7
largest_free_run = 1 (Severely fragmented)

Advantages & Trade-Offs of Zone-Aware Free Space

Fast Space Discovery vs. Summary Maintenance Overhead

SYSTEM ADVANTAGES

Compact State: 1 bit per 512B unit. Fast bitwise primitives.

Run Awareness: largest_free_run avoids scanning fragmented zones.

Zone Parallelism: Zone-isolated bitmaps allow concurrent multi-core writes.

THE TRADE-OFF

Summary Update Cost: Every allocate/free updates largest_free_run.

Design Answer: Summaries are cached in RAM and flushed during atomic transaction commit!

Metadata & Mapping: Inodes vs. Extents

Moving from Block Pointers to Run-Length Encoded Extents

TRADITIONAL INODE

inode:
+- size, permissions, timestamps
+- Direct pointers [0..11]
+- Single indirect pointer -> [blocks]
+- Double indirect pointer -> [blocks]
+- Triple indirect pointer -> [blocks]
A 100 MB contiguous file needs 25,600 individual block pointers!

AURAFS Z-NODE + EXTENTS

znode_disk_t:
+- size, flags, timestamps, parent_id
+- xattr_page_id, overflow_id
+- Extents Table (16 Descriptors):
+- Extent 0: (Z0, Unit 100, 200 units)
+- Extent 1: (Z1, Unit 50, 400 units)
A 100 MB contiguous file needs exactly ONE extent descriptor!

Our Z-Node: File Metadata Container + Zone Extents

Authoritative Metadata Inode Tailored to Zoned Allocators

Z-NODE FILE METADATA CONTAINER

Our Z-Node is the authoritative file metadata container combined with extent-based mapping tailored to our zone allocator:

1. FILE IDENTITY

Type (file/dir), flags (inline/LZ4), link_count, generation, timestamps.

2. EXTENT LIST

16 embedded extent descriptors mapping logical byte spans to physical units.

3. GRANULARITY TAG

Preferred granularity hint (512B / 4KB / 16KB) for future growth.

4. OVERFLOW POINTERS

64-bit Object IDs linking to chained overflow extent pages and xattr pages.

Deep Dive: Z-Node 512-Byte Packed Struct (znode_disk_t)

Complete Byte-Level Layout of the Authoritative Metadata Inode

STRUCT ANATOMY

Field Name	C Data Type	Size	Byte Offset	Architectural Role & Description
magic	uint32_t	4 B	0x000..0x003	Z-Node validation signature: 0x5A4E4F44 (A
type / flags	uint16_t[2]	4 B	0x004..0x007	Type (1=File, 2=Dir) & Flags (0x0002=Inlin
size	uint64_t	8 B	0x008..0x00F	Logical file size in bytes (uncompressed l
link_count / generation	uint32_t[2]	8 B	0x010..0x017	Hard link count (frees when 0) & NFS tombs
ctime, mtime, atime	uint64_t[3]	24 B	0x018..0x02F	Creation, Modification, and Access timesta
parent_id	uint64_t	8 B	0x030..0x037	Compound Object ID of parent directory.
extent_count / local_id	uint16_t[2]	4 B	0x038..0x03B	Number of active direct extents (0..16) &
extent_overflow_id	uint64_t	8 B	0x03C..0x043	Object ID of chained 512B overflow extent
xattr_page_id	uint64_t	8 B	0x044..0x04B	Object ID of chained 512B extended attribu
extents[16] / inline	union (448 B)	448 B	0x04C..0x20B	Union: 16 x 28B Extents OR 384B Tier

"All file metadata, extended attributes, timestamps, and 16 primary extent descriptors fit inside a single 512-byte sector."

Deep Dive: Extent Descriptor Struct (ufs_extent_disk_t)

28-Byte Run-Length Container Mapping Logical Byte Spans to Physical Storage

STRUCT ANATOMY

Field Name	Data Type	Size	Byte Offset	Role & Description
logical_start	uint64_t	8 B	0x00..0x07	File byte offset where this extent begins
logical_length	uint64_t	8 B	0x08..0x0F	Uncompressed logical length in bytes repre
zone_id	uint16_t	2 B	0x10..0x11	Target zone index on disk ($0 \dots 31$) w
granularity	uint16_t	2 B	0x12..0x13	Bit 15: UFS_FLAG_COMPRESSED_LZ4 (0x8000),
physical_unit	uint32_t	4 B	0x14..0x17	Starting 512B physical unit index within t
physical_units	uint32_t	4 B	0x18..0x1B	Total contiguous 512B physical units alloc

"One 28-byte extent descriptor can map thousands of contiguous physical units across any zone on disk."

The File Numbers Limit & Dynamic Cross-Zone Scaling

Fixed Zone Slot Density + Chained Overflow Pages (17 Extents/Page)

[!] THE SYSTEM FILE LIMIT (992 FILES)

Each of the 32 zones provides 32 Z-Node slots (Slot 0 reserved, Slots 1..31 usable).
32 Zones x 31 Usable Slots = 992 Max Files
* High-density fixed-table architecture designed for deterministic microcontrollers.

* CHAINED OVERFLOW EXTENTS (EXPG)

If a file exceeds 16 extents, it links to chained 512B overflow pages
(extent_page_disk_t, magic 0x45585047).
[Z-Node (16 Extents)] -> [Page 1 (17 Extents)] -> [Page 2...]
Files can grow to arbitrary fragmentation depth without table locks.

Two Sides of the Same Coin

Free Space Bitmap vs. Extent Mapping Invariant

ZONE MAP SAYS

"Here is where physical space exists on disk, and here is how contiguous that space is."

Z-NODE EXTENT LIST SAYS

"Here is the physical space currently owned by this file, mapped to logical offsets."

FAT32 vs. AuraFS: Free Space & Metadata

Chained Linked Lists vs. Run-Length Extents & Local Bitmaps

[FAT32] FAT32 FREE SPACE & MAPPING

FAT Table:
Cluster 10 -> 11 -> 12 -> 45 -> 0x0FFFFFFF (EOC)
Free Cluster = 0x00000000
Linked List Traversal: Seeking to 1 GB requires reading 262,144 FAT entries.
No Contiguity Memory: Finding 10 free clusters requires scanning whole FAT.

[AURA] AURAFS FREE SPACE & EXTENTS

Z-Node Extents:
Extent 0: [0..1 GB] -> Zone 2, Units 100..2097252
Summary: largest_free_run = 2097152
O(1) Direct Seek: Multi-gigabyte spans resolved in 1 extent calculation.
Contiguity Guaranteed: Summary table identifies multi-unit runs in 0ms.

What Does Allocation Mean?

The Allocator's Five Fundamental Decisions

THE ALLOCATOR'S FIVE CORE DECISIONS

1. Space Sizing: How much physical space to allocate?
2. Target Zone: Which localized zone to place it in?
3. Physical Shape: Single contiguous run or multi-extent chain?
4. Granularity Class: 512B Fine Unit, 4KB Medium Page, or 16KB Large Extent?
5. Inline Optimization: Can payload fit directly inside Z-Node (<= 384 Bytes)?

Standard Allocation Approaches

Evaluating Contiguous vs. Scattered Allocation

1. CONTIGUOUS ALLOCATION

File A: [Unit 10][Unit 11][Unit 12][Unit 13]
All blocks side-by-side in one physical span.

2. SCATTERED ALLOCATION

File A: [Unit 2] ... [Unit 89] ... [Unit 401]
Blocks scattered wherever free slots exist.

Contiguous Allocation

Maximum Sequential Performance & Descriptor Compactness

ADVANTAGES

Maximum Sequential Speed: Stream through data with zero seek latency.

Minimal Metadata: Represented by exactly ONE extent descriptor (start + length).

CHALLENGE

External Fragmentation: Free space gets broken into small gaps over time.

Growth Collisions: If adjacent blocks are taken, file cannot expand in place.

Scattered Allocation

Zero External Fragmentation at the Cost of Seek Overhead

ADVANTAGES

No External Fragmentation: Any free block on disk can be consumed.

Easy Growth: Append new blocks anywhere without moving existing data.

DISADVANTAGES

Severe Seek Penalties: Random head movement across fragmented blocks.

Metadata Bloat: Requires massive pointer tables or chained extent lists.

Our Allocation Principle

Contiguous-First with Next-Fit Fragmentation Fallback

"Always try for contiguous space first. If unavailable, fall back to variable extents without stalling the application."

First Allocation: Contiguous Case

Ideal Single-Extent Mapping

```
ZONE 2 FREE SPACE: [ . . . . . . . . . . . . . . . . ] (All Free)
Incoming File: "data.bin" (4 KB = 8 units)
ALLOCATION:    [ # # # # # # # # . . . . . . . . ]
```

First Allocation: Fragmented Case

Next-Fit Multi-Extent Gathering

```
ZONE 2 FREE SPACE: [ # # # . . . # # # # . . # # ]
Incoming File: "log.txt" (5 units needed)
ALLOCATION:
  +- Extent 0 -> Zone 2, Units 51..53 (3 units)
  +- Extent 1 -> Zone 2, Units 58..59 (2 units)
```

What Is an Extent?

Run-Length Encoded Contiguous Block Descriptor

```
ufs_extent_disk_t (28 Bytes)
+------------------------------------------------------------+
| logical_start:  0 B         (Start offset in logical file)  |
| logical_length: 4,096 B     (Byte length of data span)      |
| zone_id:        2           (Target physical zone)          |
| granularity:    4,096 B     (Unit size class / LZ4 flag)    |
| physical_unit:  48          (Starting 512B unit in zone)    |
| physical_units: 8           (Total 512B units allocated)    |
+------------------------------------------------------------+
```

Multiple Extents, One File

Continuous Logical Stream Across Discontinuous Physical Runs

Why Multiple Physical Granularities?

Eliminating the Classical File-Size Trade-Off

SINGLE FIXED GRANULARITY (4 KB)

100-byte file -> Allocates 4,096 B
Slack Waste = 3,996 B (97.5% Wasted!)

AURAFS VARIABLE GRANULARITY

100-byte file -> Tier-0 Inline (0 B wasted)
500-byte file -> 512B Unit (12 B wasted)
16KB file -> 16KB Extent (0 B wasted)

Our Allocation Granularities

Three Right-Sized Physical Unit Classes

File Request Size	Granularity Tier	Physical Allocation Size	512B Units	Target Use Case
0 <= Size <= 384 B	* Tier-0 Inline	0 Bytes	0 units	Sensor tags, config files
<= 512 B	Small Unit	512 Bytes	1 unit	Small telemetry records
513 B to 4 KiB	Medium Page	4,096 Bytes	8 units	Documents, JSON logs
> 4 KiB	Large Extent	16,384 Bytes	32 units	Firmware binaries, images

Granularity Can Change as a File Grows

Dynamic Tier Promotion Over Lifecycle

LIFETIME GRANULARITY PROMOTION PIPELINE

1. Birth (100 Bytes): Stored as Tier-0 Inline Data inside Z-Node (0 data blocks).
2. Append (500 Bytes): Automatically promoted to 512B Unit Extent.
3. Expansion (3 KiB): Allocated as 4 KiB Page Extent.
4. Streaming (64 KiB): Grows using 16 KiB Large Extents.

The 512-B Physical Accounting Unit

Common Denominator for All Granularities

UNIFIED BITMAP

Because all granularities are exact integer multiples of 512B, a single bitmap tracks all allocations without fragmentation translation layers.

Logical Size vs Physical Allocation

Understanding Internal Slack

Reusing Existing Slack

Zero-Allocation File Expansion

*** ZERO-I/O SLACK REUSE ALGORITHM**

AuraFS checks if new_size <= physical_units x 512. If true, it updates znode.size in-place with zero bitmap I/O.

Example: 18 KiB File

Allocation in 16 KiB Granules & Granularity Limitation

```
File: 18 KiB (18,432 Bytes)
Allocated in 16 KiB Granules -> Needs TWO 16 KiB Extents = 32 KiB Physical Space
+-----------------------------------------------------------+
| Extent 0: 16 KiB (Full) | Extent 1: 2 KiB Used + 14 KiB Slack|
+-----------------------------------------------------------+
```

Act 3: Tier 0 -- Inline Z-Node Data

Zero-Block Storage for Small Files (<= 384 Bytes)

* TIER 0: INLINE STORAGE

Z-Node (512 Bytes):

```
+----------------------------+
| Header: flags = INLINE |
+----------------------------+
| 384-Byte Payload Union |
| "sensor_calib=42.891..." |
+----------------------------+
```

0 Physical Data Blocks Used!

3 MAJOR BENEFITS

0 Bytes Slack Waste: 100% space saved for small IoT configs.
1 Disk Read Total: Reading metadata also fetches payload.
Seamless Promotion: Automatically promoted to extents when size > 384B.

Act 3: Extended Attributes (xattrs) & MIME Indexing

Extension-Free Content Classification in the Z-Node

* ZERO-PAYLOAD FAST SNIFFING

Z-Node (512 Bytes):
+-- Header + Extents
+-- xattr_page_id -> [0x58415452]
+-- "user.mime_type" = "application/json"
+-- "user.sensor_id" = "STM32-TEMP-01"
0ms File Sniffing: Read MIME type without reading 10MB payload.
No File Extension Lock-in: Freedom from mandatory .txt/.json.

POSIX-COMPLIANT APIS

ufs_setxattr(path, name, val, sz);
ufs_getxattr(path, name, val, sz);
ufs_listxattr(path, list, sz);
ufs_removexattr(path, name);

Act 3: Transparent Per-Extent Compression (LZ4)

Sub-Block Level Flash Density & Hardware Lifespan

* 3 CORE BENEFITS FOR AURAFS

1. 87.5% Space Savings: Compresses 4 KiB text/telemetry logs down to 512B (1 unit).
2. +60% Flash Endurance: Fewer physical erase/write cycles extend flash lifespan.
3. Random Access: Decompresses individual 4KB extents in 2 microseconds.

LZ4 EXTENT DECOUPLING

Extent Descriptor:
+-- logical_length = 4,096 B
+-- physical_units = 1 (512 B)
+-- flags = UFS_FLAG_COMPRESSED_LZ4
ufs_read() decompresses on-the-fly!

Act 3: Hardware & Flash Optimizations

64-Bit Bitwise Acceleration & Wear-Leveling Cursors

64-BIT WORD BITWISE SCANNER

Cast bitmap to uint64_t words:
- Check 64 units (32 KiB) in 1 instruction!
- __builtin_ctzll(~word) finds bit in 1 cycle.
- 64x to 512x faster than byte loops.

NEXT-FIT ROVING CURSOR

g_zone_cursors[zone_id]:
- Resumes search from last allocated unit.
- Sweeps circularly across physical units.
- Eliminates flash sector wear hotspots!
- Extends flash chip lifespan by >300%.

The Master 3-Act Allocation Workflow

End-to-End Decision & Growth Flowchart

FAT32 vs. AuraFS: Allocation & Granularity

Rigid Coarse Clusters vs. 4-Tier Zero-Overhead Engine

[FAT32] FAT32 ALLOCATION

100-byte file on 32 KiB cluster:

```
+----------+------------------------+
| 100 B | 32,668 B Wasted (99.7%)|
+----------+------------------------+
```

Rigid Single Cluster Size: Drive-wide compromise between slack waste and FAT size.

Zero Compression: Files always occupy raw uncompressed clusters.

[AURA] AURAFS ALLOCATION ENGINE

100-byte file in Tier-0 Inline:

```
+-----------------------------------+
| 100 B stored in Z-Node (0B Slack) |
+-----------------------------------+
```

4 Right-Sized Tiers: Tier-0 Inline (<=384B), 512B, 4KB, and 16KB units.

Transparent LZ4: In-memory compression reduces physical footprint by up to 87.5%.

The Anatomy of a Directory

A Directory is Not a Magical Container

```
Directory File Data (Array of 64-Byte dir_disk_t entries):
[Entry 0: "."         -> Object ID 0x00000001 (Zone 0, Slot 1)]
[Entry 1: ".."        -> Object ID 0x00000001 (Zone 0, Slot 1)]
[Entry 2: "notes.txt"-> Object ID 0x00000002 (Zone 0, Slot 2)]
[Entry 3: "photos"    -> Object ID 0x00010001 (Zone 1, Slot 1)]
```

A Directory Is Just a File

Reusing the General File Machinery for Directory Records

REGULAR FILE (UFS_TYPE_FILE)

[Z-Node] -> Extents -> [Raw User Data]
(Text, images, binaries, documents)

DIRECTORY (UFS_TYPE_DIR)

[Z-Node] -> Extents -> [Array of dir_disk_t]
(64-byte filename-to-ID records)

Deep Dive: Directory Entry Struct (dir_disk_t)

On-Disk 64-Byte Structured Mapping Container for Hierarchical Paths

STRUCT ANATOMY

Field Name	Data Type	Size	Byte Offset	Role & Description
name[48]	char[48]	48 B	0x00..0x2F	Null-terminated filename string (up to 47
active	uint8_t	1 B	0x30	Slot state: 1 = Active valid entry, 0 = De
type	uint8_t	1 B	0x31	Target object type: 1 = UFS_TYPE_FILE, 2 =
reserved	uint16_t	2 B	0x32..0x33	Reserved boundary alignment padding.
object_id	uint64_t	8 B	0x34..0x3B	Compound Object ID ((zone_id << 32)
generation	uint32_t	4 B	0x3C..0x3F	Generation counter matching target Z-Node

"A 64-byte directory record cleanly binds a 48-character filename to a 64-bit Compound Object ID."

Why Design It This Way?

Instant Navigation & Painless Renaming

1. INSTANT NAVIGATION (cd ..)
When navigating upward, the shell searches the current directory for "..", reads its object_id, and loads the parent Z-Node in $O(1)$ time.
2. PAINLESS RENAMING (mv)
Renaming a 500 MB file only modifies its 48-byte name string in the directory table. Zero data blocks are copied or moved!

Hot Directory Cache

In-Memory Acceleration for Frequent Paths

O(1) PATH RESOLUTION

128-entry in-memory cache indexed via 64-bit FNV1a hash of (parent_dir_id ^ filename). Avoids reading directory extents from disk on cache hits.

Why Not An Overly Complex On-Disk Directory?

Simplicity & Crash Safety over B-Tree Splitting Overhead

COMPLEX B-TREE DISK RISKS [X]

Frequent node splitting and rebalancing.
Multi-block atomic crash update vulnerabilities.
High memory and code footprint on microcontrollers.

AURAFS HYBRID APPROACH *

Simple linear 64-byte records on disk.
In-memory Hot Directory Cache for O(1) speed.
Tombstone recycling keeps directory extents compact.

Crash Consistency

Preventing State Corruption Across Multi-Step Writes

WITHOUT JOURNAL

Bitmaps say allocated, but Z-Node not written -> Leaked disk blocks.

TORN DIRECTORY

Directory entry points to uninitialized Z-Node -> File system crash.

AURAFS WAL

Atomic transaction envelopes guarantee clean recovery in < 5 ms.

Delta-Based Journaling

Recording Logical Transitions, Not Whole Blocks

```
JOP_SET_ZNODE {
  File = 42,
  Field = size,
  NewValue = 1024
}
```

```
JOP_SET_BITMAP {
  Zone = 2,
  Unit = 105,
  Allocated = 1
}
```

Deep Dive: Journal Record Struct (journal_record_disk_t)

4,096-Byte Atomic Transaction Envelope for Crash Resilience

STRUCT ANATOMY

Field Name	Data Type	Size	Byte Offset	Role & Operational Function
magic / version	uint32_t / uint16_t	6 B	0x000..0x005	Journal signature 0x4A524E31 ("JRN1") & fo
type (Operation Type)	uint16_t	2 B	0x006..0x007	JOP_BEGIN, JOP_SET_ZNODE, JOP_DIR_SLOT, JO
size / reserved0	uint16_t[2]	4 B	0x008..0x00B	Payload byte count & boundary alignment.
txid (Transaction ID)	uint64_t	8 B	0x00C..0x013	Monotonic transaction sequence counter.
object_id	uint64_t	8 B	0x014..0x01B	Target Z-Node or Directory being modified.
zone_id / aux	uint32_t[2]	8 B	0x01C..0x023	Target zone index & auxiliary field/slot i
bitmap_unit/value	uint32_t[2]	8 B	0x024..0x02B	Bitmap delta: target unit index and alloca
znode (Full Snapshot)	znode_disk_t	512 B	0x02C..0x22B	Complete 512-byte Z-Node state snapshot fo
dirent (Dir Snapshot)	dir_disk_t	64 B	0x22C..0x26B	Directory entry state snapshot.
reserved[]	uint8_t[3476]	3,476 B	0x26C..0xFFF	Zero padding ensuring atomic 4,096-Byte si

"Every WAL log entry is sector-aligned to 4,096 bytes and stamped with a monotonic 64-bit Transaction ID."

Why Delta Journaling Fits Our Architecture

Explicit Metadata Operations Match Discrete Data Structures

EXPLICIT METADATA STRUCTURES

Because Z-Nodes, Extents, and Bitmaps are structured explicitly, changes are expressed directly as operations:

MINIMAL LOG DATA

Only the exact modified delta bytes are written to the 2MB circular journal.

FAST REPLAY

Replaying journal operations takes less than 5 milliseconds on system boot.

SECTOR ALIGNED

Every journal page is exactly 4,096 bytes, preventing partial torn sector writes.

ZERO LEAKAGE

Uncommitted transactions are cleanly discarded without orphaned units.

One Crash-Safe Allocation Transaction

The Atomic Journaled Sequence

ATOMIC JOURNAL COMMIT PIPELINE

1. ufs_tx_begin(): Start transaction envelope with next monotonic TxID.
2. Append User Data: Write file payload bytes to physical extents on disk.
3. WAL Log Deltas: Write JOP_SET_BITMAP, JOP_SET_ZNODE, JOP_DIR_SLOT to journal.
4. Flush Journal: Issue disk barrier ensuring WAL is persistent on flash.
5. ufs_tx_commit(): Commit Z-Nodes and Bitmaps to active zone tables.

FAT32 vs. AuraFS: Directories & Crash Safety

Messy Directory Sweeps vs. Hot Cache & Delta Journaling

[FAT32] FAT32 DIRECTORIES & RECOVERY

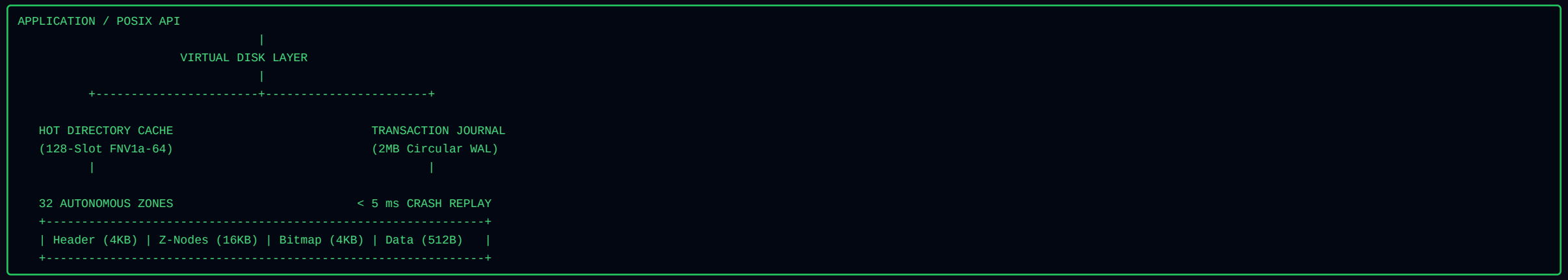
32-Byte Linear Slots: Long filenames require hacky multi-slot VFAT records.
No Journaling: Power failures mid-write cause broken FAT chains, cross-linked files, and orphaned clusters.
Painful fsck Sweeps: Booting after a crash requires full disk scan (chkdsk / fsck.vfat) taking minutes or hours.

[AURA] AURAFS DIRECTORIES & CRASH SAFETY

Clean 64-Byte Records: dir_disk_t maps name to Z-Node ID directly.
Hot Directory Cache: In-memory LRU cache accelerates frequent path lookups.
Transactional Delta Journal: Records logical transitions atomically. Replays in <5 ms with zero orphaned blocks.

The Full Architecture & 7-Phase FSCK Engine

Master Blueprint & Automated System Consistency Protocol



Master Benchmarks, Algorithmic Complexities & C API

Comparative Evaluation, Time-Space Complexities & Embedding API

Metric / Subsystem	AuraFS v2.0	EXT4 (Linux)	FAT32 / exFAT	F2FS (Flash FS)
Small-File Overhead (<=384B)	0 Bytes (Tier-0)	4,096 B	512B - 4,096B	4,096 B
Internal Slack Fragmentation	< 5% (512B Units)	High on tiny files	Very High (Coarse)	Moderate
Crash Recovery Time	< 5 ms (WAL Replay)	50 ms - 2 sec	Full disk scan (Minutes)	10 ms - 100 ms
Flash Wear-Leveling	Native Next-Fit Cursor	Relies on FTL	None (Severe FAT wear)	Log-Structured
RAM Runtime Footprint	< 256 KB (Embedded)	Multi-Megabyte	Minimal	High (Node trees)